INCH-POUND
MIL-DTL-24643/77
w/AMENDMENT 1
5 May 2020
SUPERSEDING
MIL-DTL-24643/77
29 October 2019

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +176 DEGREES FAHRENHEIT (+80 DEGREES CELSIUS), 150 VOLTS, TYPE LSC6FS, LSC6OS (INCLUDING WATERTIGHT VARIATION LSC6OSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in three variations: LSC6FS (Metallic Foil Shield Non-watertight), LSC6OS (Metallic Foil and Braid Shielded Non-watertight), LSC6OSW (Metallic Foil and Braid Shielded Watertight).

Construction

First - Copper conductor, solid, uncoated copper, in accordance with ASTM B3, 23 AWG nominal. Conductor diameter may vary from nominal as necessary in order for the completed cable to meet the specified electrical requirements.

Second - Solid or foamed polyolefin insulation. Wall thickness as necessary, in order for the completed cable to comply the electrical requirements specified herein. The maximum insulation diameter shall not exceed 0.051 inch.

Third - Two insulated conductors, twisted together with an appropriate lay length for each pair such that the completed cable complies with the electrical requirements specified herein. Wire identification codes shall be in compliance with ANSI/TIA-568.2 as follows:

Pair 1 – white/blue and blue
Pair 2 – white/orange and orange
Pair 3 – white/green and green
Pair 4 – white/brown and brown

Optional identification codes for cables with tightly twisted pairs less than 1.5 inches (38 millimeters) are as follows.

Pair 1 – white and blue
Pair 2 – white and orange
Pair 3 – white and green
Pair 4 – white and brown

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Fourth - Four pairs shall be cabled together with a lay as required to comply with electrical requirements specified herein. See table I. Cabling sequence shall be consecutive. Fillers or other means may be used as necessary to form a firm, well-rounded cable. For watertight variants (type LSC6OSW only), filler material shall comply with the requirements of 3.3.9 of MIL-DTL-24643.

Construction Variant for Foil Shield Type LSC6FS

Fifth - An optional binder.

Sixth - Copper drain wire, tin-coated, 24 AWG, solid or stranded, ASTM B286.

Seventh - Metallic foil/polyester tape (metallic foil side facing in and in contact with the drain wire). Tape may be longitudinally or helically applied, with a minimum overlap of 10 percent.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Minimum thickness of 0.040 inch.

NOTE 1: Manufacturer's identification tape is required (see 3.4.9).

Construction Variant for Foil and Braided Shield Types LSC6OS and LSC6OSW

Fifth - An optional binder.

Sixth - Metallic foil/polyester tape (metallic foil side facing out and in contact with the braid).

Seventh - Braided shield of 36 AWG or 34 AWG tin-coated copper.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Minimum wall thickness of 0.040 inch.

NOTE 2: Manufacturer's identification tape is required (see 3.4.9).

TABLE I. Details.

Military part no. M24643/77	Type and size	Number of pairs	Overall diameter (inch)	
			min.	max.
-01UO	LSC6FS-4	4	0.330	0.380
-02UO	LSC6OS-4	4	0.350	0.400
-03UO	LSC6OSW-4	4	0.380	0.440

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/100 meters at 68 °F [20 °C], max.)	9.38
Conductor resistance unbalance (percent, max.)	4
Insulation resistance (megohms/1,000 meters, min.)	Not applicable

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Jacket flaws	No failure
Conductor and shield continuity	No failure
Voltage withstand:	
Conductor to conductor	1.5 kVAC or 2.5 kVDC
Conductor to shield	1.5 kVAC or 2.5 kVDC

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance, nominal at 1 kilohertz (kHZ) (nf/100 meters, max.)	5.6
Pair to ground capacitance unbalance (max. at 1 kHz) (pF/100 meters)	330
Characteristic impedance (ohms, nominal) (not a requirement)	100
Watertightness (type LSC6OSW only)	No failure
Swept electrical parameters	See table II

TABLE II. Swept electrical parameters.

Frequency (MHz)	Return loss (dB/100 m, min.)	Insertion loss (dB/100 m, max.)	NEXT (dB/100 m, min.)	PSNEXT (dB/100 m, min.)	ACRF (dB/100 m, min.)	PSACRF (dB/100 m, min.)
1.0	20.0	2.1	74.3	72.3	67.8	64.8
4.0	23.0	3.8	65.3	63.3	55.8	52.8
8.0	24.5	5.3	60.8	58.8	49.7	46.7
10.0	25.0	5.9	59.3	57.3	47.8	44.8
16.0	25.0	7.5	56.2	54.2	43.7	40.7
20.0	25.0	8.4	54.8	52.8	41.8	38.8
25.0	24.3	9.4	53.3	51.3	39.8	36.8
31.25	23.6	10.5	51.9	49.9	37.9	34.9
62.5	21.5	15.0	47.4	45.4	31.9	28.9
100.0	20.1	19.1	44.3	42.3	27.8	24.8
200.0	18.0	27.6	39.8	37.8	21.8	18.8
250.0	17.3	31.1	38.3	36.3	19.8	16.8
300.0	16.8	34.3	37.1	35.1	18.3	15.3
400.0	15.9	40.1	35.3	33.3	15.8	12.8
500.0	15.2	45.3	33.8	31.8	13.8	10.8

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TABLE II. Swept electrical parameters – Continued.

Frequency (MHz)	TCL (dB, min.)	ELTCTL (dB/100 m, min.)	Propagation delay (nS/100 m, nom.)	Delay skew (nS/100 m, max.)
1.0	40.0	35.0	570	45
4.0	40.0	23.0	552	45
8.0	40.0	16.9	547	45
10.0	40.0	15.0	545	45
16.0	38.0	10.9	543	45
20.0	37.0	9.0	542	45
25.0	36.0	7.0	541	45
30.0	n/a	5.5	n/a	n/a
31.25	35.1	n/a	540	45
62.5	32.0	n/a	539	45
100.0	30.0	n/a	538	45
200.0	27.0	n/a	537	45
250.0	26.0	n/a	536	45
300.0	25.2	n/a	536	45
400.0	24	n/a	536	45
500.0	23	n/a	536	45
NOTE: Values are reference points. Compliance shall be determined utilizing the formulas in ANSI/TIA-568.2.				

Group B:

Drip 203±1.8 °F (95±1 °C) (type LSC6OSW only)	Zero
Physical (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	450
Elongation (percent, min.)	75
Jacket	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	Not applicable
Jacket at 392 °F (200 °C)	50

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Group C:

Physicals (aged) air oven	
Insulation (212 °F [100 °C] for 48 hours)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (276.8 °F [136 °C] for 168 hours)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Cable fill (sealant) removability (type LSC6OSW only)	No failure
Heat distortion (percent of unaged, max.)	30
Shield (conformance to shield material and construction) (types LSC6FS and LSC6OSW only)	No failure
Surface transfer impedance (types LSC6OS and LSC6OSW only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Permanence of printing (jacket) (cycles, min.)	125

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; all Groups A, B, C, and D; plus the following:

Aging and compatibility (cable) 203±3.6 °F (95±2 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	8
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	55
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Stress resistance	No failure
Temperature humidity (type LSC6OSW only)	No failure
Electrical moisture absorption	Not applicable

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSC6FS, LSC6OS, and LSC6OSW	500

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2020-014)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

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